

Week - 05 - Classes, Graphics\Lecture_10_code_1.py

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1  '''
2  This code demonstrates associations between classes via
3  Dependency injection mechanism. The fire method of class Canon depends upon Projectile
4  object.
5  '''
6  import math
7  import time
8
9  G = 9.82
10 AIR_DRAG = 0.4
11
12 class Projectile:
13     def __init__(self, ID, size, energy, x_pos=0, y_pos=0):
14         self.id = ID
15         self.size = size
16         self.energy = energy
17         self.x_pos = x_pos
18         self.y_pos = y_pos
19
20     def get_projectile_data(self):
21         # A getter function
22         return (f'id {self.id}, size {self.size}, energy {self.energy} and position
23         ({self.x_pos}, {self.y_pos})')
24
25 class Canon:
26     def __init__(self, cid, length, type):
27         self.cid = cid
28         self.length = length
29         self.type = type
30
31     def fire(self, bomb, velocity, theta):
32         x_vel = velocity * math.cos(theta)
33         y_vel = velocity * math.sin(theta)
34         dt = 0.02
35
36         while bomb.y_pos >= 0:
37             bomb.x_pos += x_vel * dt
38             bomb.y_pos += y_vel * dt
39             x_vel -= AIR_DRAG * dt
40             y_vel -= G * dt
41             time.sleep(0.2)
42
43             animate(bomb.x_pos, bomb.y_pos)
44
45 # Helper function to display trajectory of our projectile
46 def animate(x, y):
47     width, height = 60, 30
48     scale = 100
49     screen = [[ " " for _ in range(width)] for _ in range(height)]
50
51     column = round(x * scale)
```

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51     row = height - 1 - round(y * scale)
52     if 0 <= column < width and 0 <= row < height:
53         screen[row][column] = "*"
54
55     print("\033[2J\033[H", end="")
56     print("\n".join("".join(line) for line in screen), flush=True)
57     time.sleep(0.1)
58
59 bomb_1 = Projectile('B101', 20, 15)
60 gun_1 = Canon('Gun-13', 20, 'Ground-based muzzle')
61
62 # We pass Projectile object (bomb_1) to canon object (gun_1)
63 gun_1.fire(bomb_1, 2, math.radians(50))
```